Gonzalo Díaz López

Software Engineer (C++, Python) · PhD in Physics

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Summary

Software engineer (**C++**, **Python**) with a PhD in physics and eight years of building scientific software. Main developer of a public data-processing framework; earlier rewrote an experiment's core software. Adapts quickly to new technical domains across three scientific projects, and has taught at university.

Experience

Software Research Engineer

CNRS, LPNHE / Sorbonne Université, Paris — *Apr 2025 – Present*

- Lead development of **Allen Core**, a public **C++** framework for processing scientific data on **CPUs and GPUs**, derived from the real-time system of CERN's LHCb experiment (gitlab.cern.ch/allen-core).
- Build a demonstrator that proves the framework is generic and usable by other experiments: it runs another project's complete algorithm chain, on either GPU or CPU.
- Package the framework for others to adopt: installable build, **containers** and **continuous integration**.

Postdoctoral Researcher

CNRS, LPNHE / Sorbonne Université, Paris — *Apr 2023 – Mar 2025*

- Rewrote, single-handedly, Hyper-Kamiokande's software that turns detector signals into physics events, from deprecated **C++** to **Python**: faster, easier to debug, and now the base for PhD students' projects.
- Ran simulation productions on **high-performance computing (HPC)** clusters to test and validate the new software.
- Supervised a Master's student through a six-month project.

PhD Researcher

IGFAE, Universidade de Santiago de Compostela — *Sep 2018 – Feb 2023*

- Made the NEXT-100 detector simulation 1,000× faster by replacing repeated calculations with a lookup table, so a dataset equal to 20 years of detector operation became feasible.
- Implemented that simulation in **C++** and contributed **Python** software for data reconstruction and analysis.
- Implemented **machine-learning** methods that improved how detector events are classified.

Skills

- **Programming:** C++, Python, Bash, CUDA (basic)
- **Engineering:** object-oriented design, multi-threading, memory debugging (gdb), code review
- **Build & delivery:** Git, CMake, GitLab CI/CD, Apptainer, Docker, pytest
- **Computing:** Linux, HPC clusters (Slurm), GPU computing
- **Data & machine learning:** NumPy, Pandas, HDF5, Matplotlib, scikit-learn, PyTorch, Keras

Education

PhD in Particle Physics
Universidade de Santiago de Compostela *2018 – 2023*

Thesis: *Sensitivity of the NEXT-100 detector to the search of neutrinoless double beta decay*.

MSc in Nuclear and Particle Physics
Universidade de Santiago de Compostela *2017 – 2018*

Thesis: *Double beta identification studies in the NEW detector with neural networks*.

BSc in Physics
Universidade de Santiago de Compostela *2013 – 2017*

Thesis: *Critical Phenomena and Gravitational Collapse*.

Teaching

- **Numerical Physics**, Master's level (36 h) — Université Paris Cité, 2024–25.
- **Multivariate analysis** (Mathematical Methods III) and **physics lab sessions**, Bachelor's level (20 h/year each) — Universidade de Santiago de Compostela, 2019–21.

Publications

- *Energy calibration of the NEXT-White detector with 1% resolution near Qββ of ^{136}Xe*, JHEP 10 (2019) 230 (co-author).
- *Demonstration of the event identification capabilities of the NEXT-White detector*, JHEP 10 (2019) 052 (co-author).

Languages

Spanish (native) · Galician (native) · English (professional) · French (intermediate)